# Week 5 Progress Report

**ECE501: Digital Image Processing** | **GROUP 19** | October 18, 2026

**Project Topic:** Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

**Deliverable:** Two-class (defect-free + defective) LBP and GLCM detectors, proposed extensions, and an extended evaluation plan


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## Objectives of the Week

- Apply our own variation to the Week 4 LBP and GLCM baselines (where defects were not present on the fabric).
- Train both of the detectors to normal and defect fabric and then each window will be compared to a normal model and a defect model.
- Make additional suggestions for further features, pre and post processing for both methods.
- Outline other criteria for evaluation other than Accuracy, Precision and Recall.

## Work Done This Week

### A. Dataset and training setup

- Dataset: AITEX Fabric Image Database (AFID) [3] type of fabric 03 and pixel level GT masks.
- The detectors are also self-learning in the presence of faulty images. A window is considered to be defective if the mask covers at least a portion of the pixels of the window, otherwise normal.
- Splits are done on an image basis. Only on the validation images, thresholds are selected. The 12 “defective” images used solely for calibration in Week 4 are now training images and so all test results are to be made using other images.
- Team observation: For running we found that introducing faulty pictures in the team increased the separation between faulty and normal windows over the baseline of

defect-free picture for both LBP and GLCM. Numbers that are held out are written in the Table 2.

### B. LBP method

**Implemented: two-class LBP.**

- Two reference histograms: M_N (defect free windows) and M_D (defective windows). Scales (P, w) = (8, 3), (16, 5), (24, 7); 16 x 16 windows, step 4.
- Each window histogram S gets a log-likelihood ratio: $\Lambda(S) = \Sigma\ S_k \cdot \log(M_{D,k} / M_{N,k})$. Positive score indicates the window is more towards the defect model. To prevent log(0), a small constant is added to each bin.
- There is still the 30% pixel vote, and the two flagged windows.The 30% pixel vote and the minimum of two flagged windows remain as in Week 4.

**Proposed LBP variations:**

| Variation | What changes | Problem it targets |
|---|---|---|
| Rotation-invariant uniform LBP with local variance (LBP/VAR) | Joint histogram of pattern and local contrast | Weak-contrast thin defects |
| Illumination normalisation | Per-image mean/std normalisation or CLAHE before LBP | Brightness mismatch |
| Edge-band handling | Ignore the border band or give it its own reference | Upper-edge band false alarms |
| Smaller and thin windows | 8 x 8, 8 x 32 and 32 x 8 windows, coarse-to-fine | Defects smaller than 16 x 16 |
| Histogram distance | Compare log-likelihood ratio, chi-square, histogram intersection | Threshold sensitivity |
| Classifier on histograms | SVM or Random Forest on concatenated 3-scale histograms | Unequal usefulness of scales; (8,3) flagged nothing |
| Scale weighting | Weight each scale by validation AUC | Same |
| Post-processing | Morphological opening/closing, minimum region area | Scattered tiny regions |

### C. GLCM method

**Implemented: two-class GLCM.**

- The features are not changed: contrast, dissimilarity, homogeneity, energy, correlation at 1, 2, 0, 45, 90, 135 degree (average), 16 gray levels. The defect free and buggy patches are now in the standardised feature bank.
- For each test patch, d_N, distance to nearest defect-free patch, and d_D, distance to nearest defect patch are recorded. Score: $R(x) = d_N / (d_D + \varepsilon)$. The patch is marked if R exceeds a threshold that was selected by the validation.
- A ratio is not based on one distance, but rather on some degree of "closeness" to each class, and we don't expect this to be as sensitive as in Week 4. This has yet to be quantified.

**Proposed GLCM variations:**

| Variation | What changes | Problem it targets |
|---|---|---|
| More Haralick features | As an example, the values for the first four are defined in the corresponding arguments.For example, entropy, cluster shade/prominence, sum average/variance and max probability are all passed as arguments. | Subtle defects |
| Feature selection | New methods: Fisher score or ANOVA ranking, correlation filter, optional PCA | Certain features are repeated, runs are slow. |
| Direction-aware features | Do not combine angles with each other, or include an anisotropy feature (0° vs 90° contrast). | When thin horizontal/vertical defects are hidden by averaging.If thin horizontal/vertical defects are hidden by averaging. |
| Local normalisation | Normalise data using CLAHE and quantise afterwards. | Brightness mismatch |
| Better distance | The mean of the k nearest patches or Mahalanobis with Ledoit-Wolf shrinkage. | Single-neighbour noise, covariance problem from Week 3 |
| Learned classifier | Standardised features with SVM or Random Forest. | Hand-set threshold |
| Multi-size patches | 16 x 16 and 32 x 32 combined | The size of the hole is less than one patch |

| Variation | What changes | Problem it targets |
|---|---|---|
| Faster GLCM | 8 gray levels, vectorised, parallel patches | Run time |
| Post-processing | Same as LBP | Scattered false patches |

### D. Fusion and fair comparison

**Proposed LBP + GLCM fusion** It is proposed that the local micro-patterns are captured by Local Binary Pattern (LBP) and the gray-level statistics are captured by Gray Level Co-occurrence Matrix (GLCM), which may miss different micro-patterns.

- Score-level: Normalize to a score range from 0 to 1 at the score level and multiply by a weight determined by the validation.
- Mask-level: And to eliminate false alarms, or to increase recall.
- Feature-level:combine the features of LBP and GLCM into one classifier.
- Hybrid rule: if either two-class score or normal-only score is high, flag the window, to maintain high sensitivity for the unseen type of defects.

**Comparison conditions:** The same pre-processing, 16 x 16 window grid (step 4), the same train/validation/test split, thresholds based on the validation images using each method's score, the same metrics (Table 3), and the same machine used for the run times were employed for the comparison.

### D2. Week 6 additions backed by published work

Each line links to a paper in which the concept was tested. The numbers mentioned in those papers are theirs and hence the results that we report will differ from those papers. All of the rows are proposed.

| Addition | Published support | What we will do |
|---|---|---|
| Conditioned TPR (cTPR) | Zhou et al. [7] define cTPR as the recall rate when there are no false alarms. More specifically, a ResNet achieved a 93.5% recall, 83.6% precision, yet a cTPR of ~ 0.217 compared to that of their Gabor one-class model, which was 0.895. | Report cTPR for LBP, GLCM and Gabor beside the Accuracy, Precision and Recall. |
| Gabor filter bank | [7] use 90 filters: wavelength $\lambda = 2m$ (m = 2 to 6), 10° orientation steps, $\sigma = 0.56\lambda$. More filters raised cTPR but slowed processing. Kumar and Pang [6] also use banks of scales and orientations. | Compare full bank and smaller bank and make a plot of cTPR versus run time. |
| Patch-size sweep | They used a patch size of 32, with cTPR increasing then decreasing with increasing patch size. | Run 16, 32 and 64 for LBP, GLCM and Gabor. |

| Addition | Published support | What we will do |
|---|---|---|
| Gabor + PCA | Bissi et al. [8]: 98.8% detection rate and 0.20-0.37% false alarms on TILDA images. [7] state that if there is too little information stored there, it will be lost. | Select number of components using validation cTPR. |
| One tuned Gabor filter | Jing et al. [9] tunes one filter using one genetic algorithm that is applied to one type of fabric texture. As noted by [7] it can't be transferred to other fabrics. | Grid search for the values of $\lambda$ and $\theta$ with the validation images of fabric 03 (our simpler solution to the genetic algorithm). |
| One-class against two-class | [7]: a nearest-neighbour distance-ratio score with threshold $\tau = 1.05$ which does not require any defective samples. | Compare to our 2 class ratio on the same split, this is also a test of hybrid rule. |
| Rotation robustness | In [7] test images were rotated by 0.02 to 5 degrees and the cTPR decreased with the angle. | Repeat this for LBP, GLCM and Gabor and graph out the cTPR as a function of the angle. |
| Gabor detects, LBP/GLCM names the type | A Gabor filter was employed to detect and LBP and Tamura features were employed to classify a study [10] and a review [11] reports that Gabor + LBP localised tiny defects better in an ablation. Only abstracts were read read. There are 12 types of defects [12] in AFID. | Results will be calculated later |
| Published AITEX reference | A survey [12] reports the AUROC of published methods on AITEX and a deep model [13] reports average AUROC of 98.4% (image) and 98.1% (pixel) over AITEX and Carpet-MVTecAD. | Besides these, add a small table of our ROC-AUC, with different methods and protocols. |

**E. Changes from Week 4**

| | Week 4 | Week 5 |
|---|---|---|
| Training data | Defect-free only | Defect-free + defective |
| Score | The distance from a model that is considered normal. | Normal-vs-defect ratio |
| Threshold | Largest leave one out score (LBP), margin multiplier (GLCM) | Chosen on validation images |

| | Week 4 | Week 5 |
|---|---|---|
| | Selected on images while validating | |
| Evaluation | A total of 12 images to be calibrated. | Images that are thrown out due to being held out + Table 3 metrics |

Gabor features, planned for Week 5, move to Week 6 because this week went to the two-class variations and evaluation design.

## Plan for Next Week (Week 6)

- Apply the suggested LBP changes, one by one, to the two-class LBP: illumination normalisation, edge-band handling, thin windows and post processing.
- Add the proposed gLCM variations: extra Haralick features – selection & direction-aware features, local normalisation, k-nearest & shrinkage-Mahalanobis scoring.
- Construct Gabor, and test Gabor fusion with LBP or PCA or one tuned filter (Gabor + sweep of patch size) and Gabor fusion with hybrid rule or one-class comparison.
- Perform rotation test and add the published-AITEX comparison.
- Record the runtimes for all methods, save masks and overlays using one window grid.
- Complete Tables 2 & 3 with cross-validation over images and recall, including cTPR and per-defect-type.

## Conclusion

This week LBP and GLCM were extended to train on defect-free and defective fabric and were scored against a normal model and defect model, which was found to be able to separate the two classes better than the Week 4 baselines in our experiments. There are still numbers to be held out and open thin defects, threshold sensitivity and edge/lighting false alarms. We'll make the most promising changes next week, as well as incorporate Gabor features and test our changes on held-out images.

## References

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